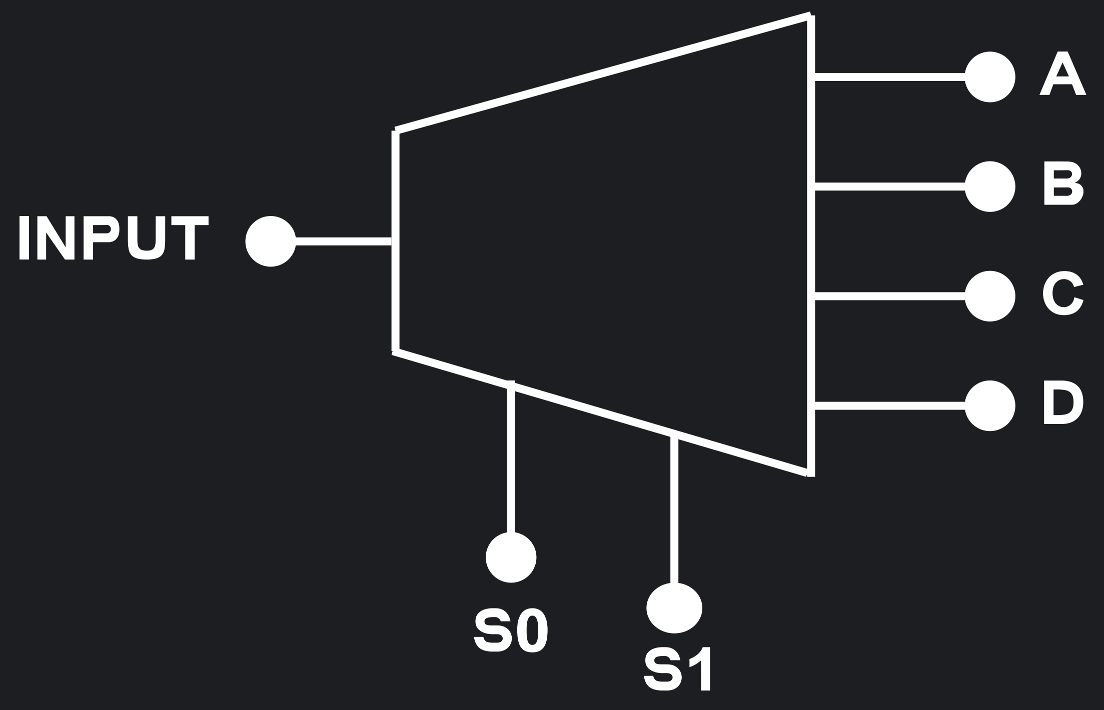
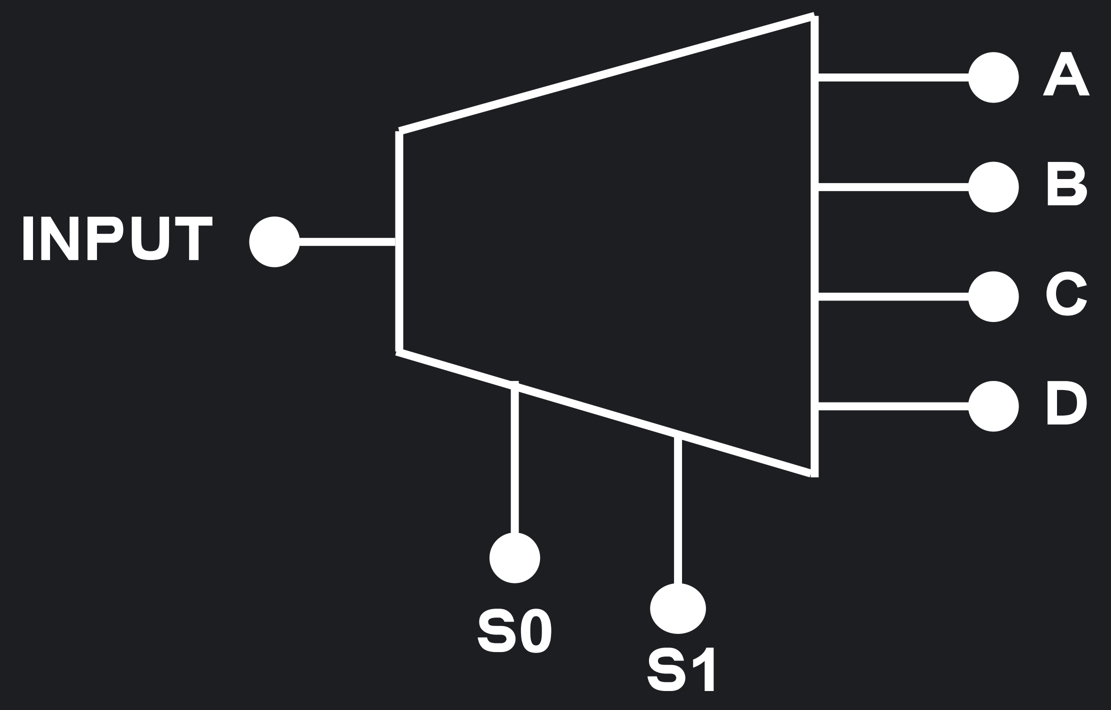
DEMULTIPLEXER 1 TO 4



What is a DEMULTIPLEXER?

A Demultiplexer (DEMUX) is a basic combinational logic element that takes a single input signal and routes it to one of several outputs based on Selector inputs. The selected output is in a HIGH (1) state if the input is HIGH, while all other non-selected outputs remain in a LOW (0) state.

Truth Table



S1	S0	Y
0	0	A
0	1	B
1	0	C
1	1	D

Function of the DEMULTIPLEXER

A Demultiplexer performs a logical data distribution operation from a single input to multiple outputs.

The main function of a Demultiplexer is to serve as a digital distribution switch that routes a single input (Data In) to one of several outputs (A, B, C, D) according to this rule:

Output A = Data In when Selectors S1_S0 = 00.

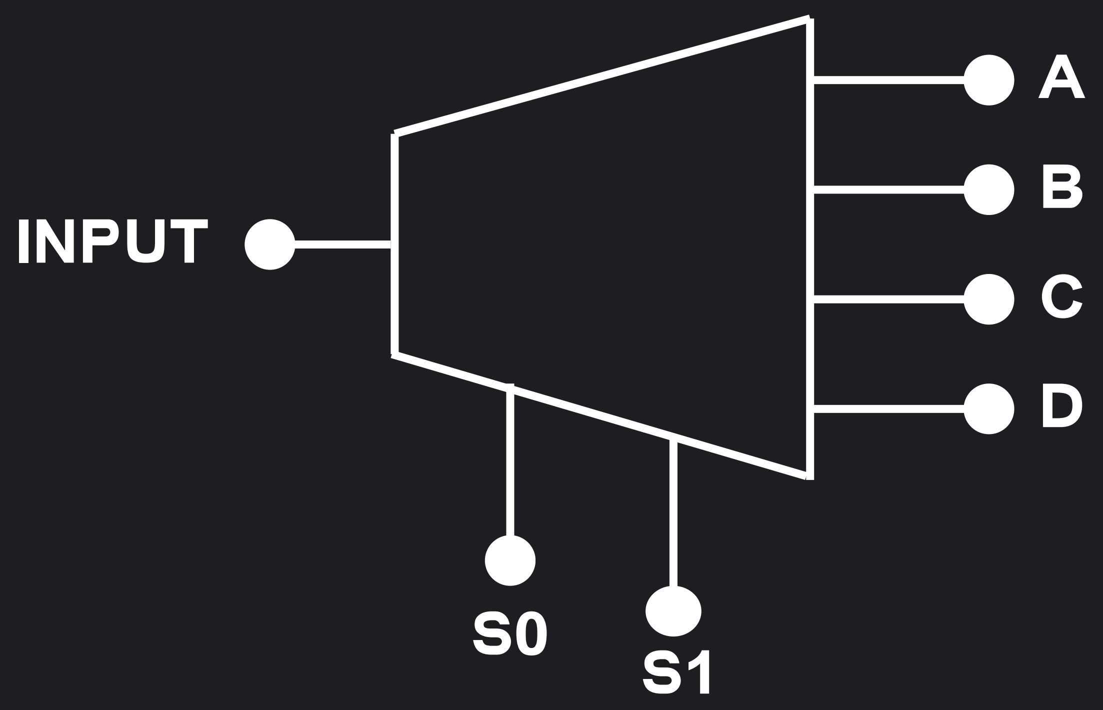
Output B = Data In when Selectors S1_S0 = 01.

Output C = Data In when Selectors S1_S0 = 10.

Output D = Data In when Selectors S1_S0 = 11.

Q\ how to connect a DEMULTIPLEXER, write the Truth Table.

Sol:



S1	S0	Y
0	0	A
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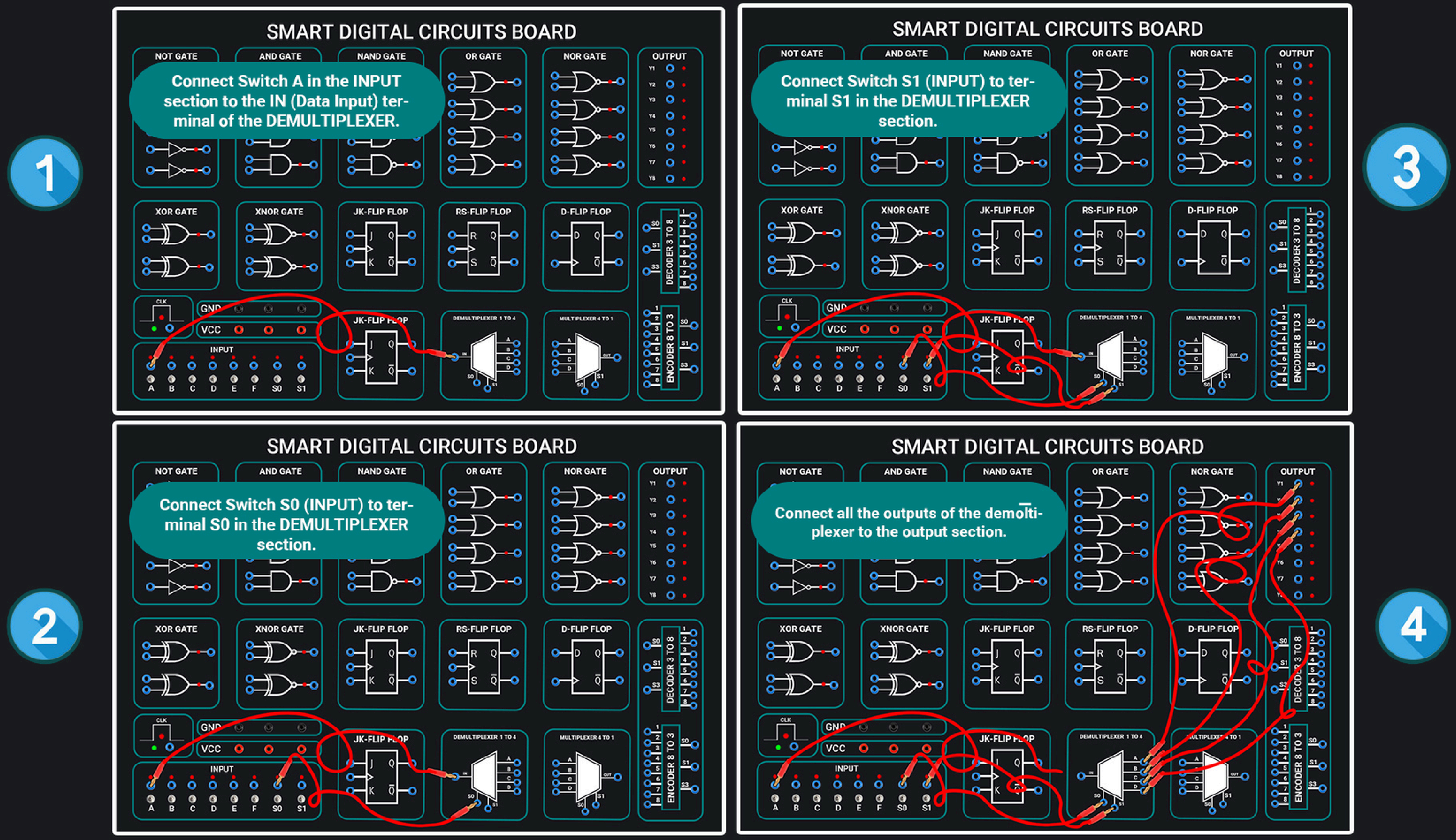
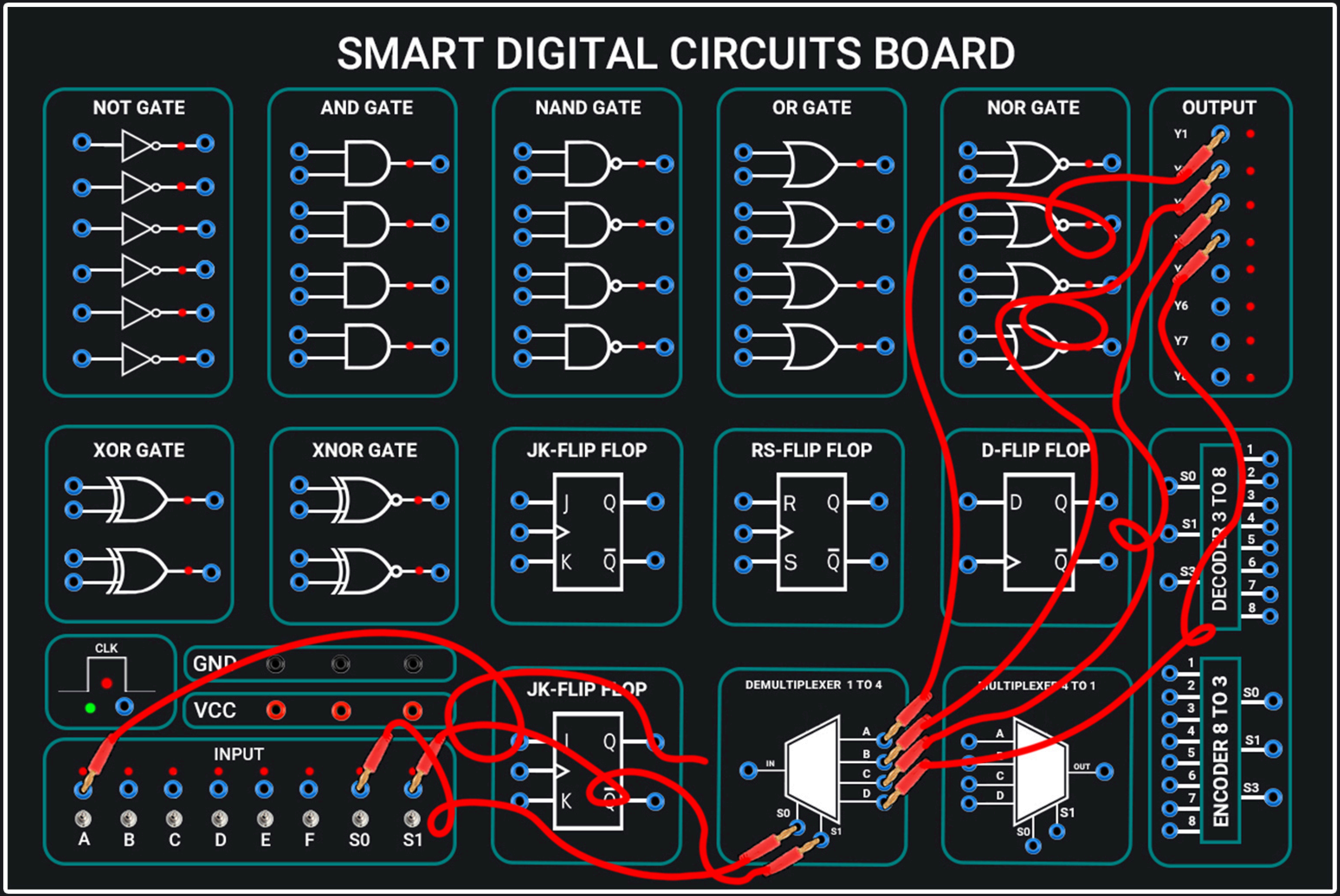
1	1	D



1
SMART DIGITAL CIRCUITS BOARD
Connect Switch A in the INPUT section to the IN (Data Input) terminal of the DEMULTIPLEXER.
2
SMART DIGITAL CIRCUITS BOARD
Connect Switch S0 (INPUT) to terminal S0 in the DEMULTIPLEXER section.
3
SMART DIGITAL CIRCUITS BOARD
Connect Switch S1 (INPUT) to terminal S1 in the DEMULTIPLEXER section.
4
SMART DIGITAL CIRCUITS BOARD
Connect all the outputs of the demolti- plexer to the output section.


SMART DIGITAL CIRCUITS BOARD
NOT GATE
AND GATE
NAND GATE
OR GATE
NOR GATE
OUTPUT
Y1
Y6
Y7
XOR GATE
XNOR GATE
JK-FLIP FLOP
RS-FLIP FLOP
R
S
Q
D-FLIP FLOP
D
DECODER 3 TO 8
S0
S1
S3
1
2
3
4
5
6
7
8
CLK
GND
VCC
INPUT
A
B
C
D
E
F
S0
S1
DEMULTIPLEXER 1 TO 4
IN
ENCODER 8 TO 3